

Certified Cognitive Glue: Value-Function Prices for Auditable Cooperation in Multi-Agent Systems

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Abstract

Multi-agent learning systems are increasingly asked to coordinate under shared norms: do not deplete a commons, repair harms, compensate costly prosocial work, and respect human interventions. Existing approaches can learn such behavior empirically, but a trained group can still fail on rare trajectories, under role changes, or when incentives create exploitable stress-harvesting cycles. We propose *certified cognitive glue*, a verification-first architecture that turns a formal progress certificate into the public coordination signal used by the agents themselves. The core object is a bounded stopped-pair supermartingale certificate for an omega-regular cooperative specification. Instead of designing separate price-like signals for scarcity, urgency, repair debt, or credit, we publish sparse interface prices derived from the certificate and its certified one-step drift. Agents may train with potential-based shaping using $\Phi = -W$, but the checker remains authoritative: a norm becomes active only after it compiles to a deterministic monitor, satisfies a bounded-repair admission gate, and passes a stopped-process supermartingale check. We prove harvest bounds for two distinct mechanisms: discounted potential shaping, which is policy-invariant for training rewards, and signed, attributed, solvency-constrained ledger accounting, which prevents cycles, resets, bankruptcy/default, or governance rebasing from creating positive net extractable surplus. We also prove a linear small-gain composition theorem under explicit interface-completeness assumptions and give an affine-disturbance variant for bounded-risk deployments. The result is a research program for producing public, auditable cooperation certificates rather than relying on empirical cooperation alone.

1 Introduction

A central problem for cooperative artificial intelligence is not merely whether a group of agents can learn to cooperate in a benchmark. The harder question is whether the cooperation is auditable: can an outside verifier inspect the agents, their environment, and their rules, and obtain a certificate that the group will keep its cooperative obligations under stated assumptions?

Consider a simple commons-cleanup game. Several agents harvest apples. Pollution accumulates in a river. Some agents can clean the river at personal cost; others benefit from the cleaner’s work. Useful social rules are easy to state informally: do not steal, do not overharvest, clean when pollution is high, compensate cleaners, and do not retaliate after repair. Yet informal rules do not provide a guarantee. An agent may learn to appear cooperative while exploiting rare states; a novice may fail to infer an obligation that is rarely observed; a price or reward signal may encourage agents to create crises and then collect reward for solving them.

This paper proposes a verification-centered architecture for such settings. The main idea is simple: if a formal certificate already measures how much cooperative trouble remains, use that certificate as the public coordination signal. In economics, prices coordinate decentralized plans by summarizing relative scarcity. In our setting, a supermartingale certificate W summarizes the remaining distance to discharging a formal obligation. We therefore treat W as a kind of certified price: high W means unresolved repair debt; a guaranteed expected decrease in W means progress; a realized checker-verified signed change can be posted as credit or debt.

This gives the architecture its name: certified cognitive glue. A cognitive glue is a shared signal or interface that lets autonomous parts form compatible plans. A certificate is a proof object showing that a stochastic process satisfies a temporal specification. Certified cognitive glue is the identification of these two objects: the proof object generates the coordination signal.

The proposed architecture is meant to be accessible to both machine-learning and economics audiences. For machine learning, the contribution is a way to connect norm learning and reinforcement learning to post hoc formal certificates. For economics and institutional design, the contribution is a formal analogue of a price system whose prices are not hand-designed but derived from a proof of obligation discharge.

Contributions. We make seven contributions.

1. We define certified cognitive glue for Markov games with formal cooperative norms.
2. We show how a fixed decentralized joint policy induces a Markov chain over an augmented monitor state, so that Markov-chain proof rules apply.
3. We replace the informal Streett proof sketch with a stopped-pair supermartingale theorem that does not require drift assumptions on discharge states B_i .
4. We define certificate-derived glue prices: repair debt, urgency, scarcity, and verified credit are functionals of W and its certified drift.
5. We prove harvest lemmas for certificate-derived rewards: discounted potential shaping is policy-invariant and bounded, while transferable public credit must use signed, attributed, solvency-constrained ledger accounting so that cycles, resets, default, and cross-agent stress harvesting do not create positive net extractable surplus.
6. We prove a linear small-gain composition theorem under declared interface-completeness assumptions and add an affine-disturbance variant for bounded-risk cases.
7. We give a theorem-gated experimental and implementation plan, including human deviation classes, permissioned glue hijacking with ledger rebasing, missing-market counterexamples, and least-forgetting abstraction refinement.

The paper is deliberately conservative about what is proved. Abstraction transfer, automatic discovery of the true interface graph, broad nonlinear composition, and strategic model-poisoning robustness remain theorem obligations rather than completed results. The small-gain result proved here covers the declared linear interface-budget case only. The intended artifact is a public certificate package: monitor, policy, value tables or bounds, glue map, deviation class, governance rebasing log, attribution ledger, and checker output. Section 7 gives a minimal two-state monitor example showing how the stopped certificate, signed ledger, attribution rule, and governance rebasing fit together.

2 Background in Plain Terms

2.1 Temporal obligations and progress certificates

An omega-regular specification is a formal way to express long-run behavioral requirements. It can encode rules such as “whenever a violation occurs, repair eventually follows” or “if pollution is repeatedly high, cleanup repeatedly occurs.” Such specifications can be compiled into automata with acceptance conditions. We use Streett pairs (A_i, B_i) . Intuitively, A_i is a recurring bad or pending condition, and B_i is the corresponding repair or discharge condition.

A Streett condition is satisfied when, for every pair, infinite recurrence of A_i implies infinite recurrence of B_i :

$$X_t \in A_i \text{ i.o.} \implies X_t \in B_i \text{ i.o..}$$

Thus an infinite alternation between A_i and B_i is not a violation. The violating behavior is visiting A_i infinitely often after B_i has stopped occurring. This distinction matters for the proof rule below: we require progress before the next B_i hit, not a drift condition while the process is inside B_i itself.

Recent work by Abate et al. [2] shows that, under suitable reward designs, the value function of a policy can encode such a certificate. This is important because value functions are standard objects in reinforcement learning, while supermartingale certificates are standard objects in stochastic verification.

2.2 Norm learning in Markov games

Norms are social rules. Some are prohibitions: do not steal. Others are obligations: if you are responsible for cleaning and the river is polluted, clean within a deadline. Oldenburg and Zhi-Xuan [3] model such norms in Markov games and show that agents can infer and sustain shared rules through approximately Bayesian rule induction. That work supplies a natural synthesis layer: agents can learn candidate norms from behavior and plan under uncertain norm beliefs.

Our architecture separates learning from certification. Norm learning may propose a rule. Only a certified rule contributes to exported guarantees.

2.3 Cognitive glue and prices

Levin and Lyons [4] argue that prices act as a cognitive glue for economies. A price system helps autonomous agents form mutually compatible plans by summarizing relative scarcities and by making other agents’ constraints locally visible. Prices also support stress sharing, reward sharing, and credit assignment.

This paper uses the economic idea in a formal verification setting. The public glue signal should not be an arbitrary reward signal. It should report certificate-relevant scarcity: how much cooperative obligation remains unresolved, how urgent the repair is, and which accountable interface reduced verified debt.

3 Formal Setting

3.1 Markov game and certified state

Let there be n agents. A Markov game consists of a state space S , action spaces $\mathcal{A}_1, \dots, \mathcal{A}_n$, a transition kernel

$$P(s' \mid s, a_1, \dots, a_n),$$

and local reward functions R_i . A joint policy profile is $\pi = (\pi_1, \dots, \pi_n)$.

For certification we use an augmented state

$$x = (s, m, c_1, \dots, c_n, \zeta_1, \dots, \zeta_n).$$

Here s is the physical state, m is the deterministic monitor state for the temporal specification, c_i is agent i 's finite norm-state transducer, and ζ_i is finite private controller memory. The monitor m includes timers, active obligations, repair windows, and Streett acceptance state. Any variable needed to define A_i , B_i , or the monitor transition belongs in m .

Definition 1 (Certified process). *For a fixed joint policy π , the certified process is the Markov chain over augmented states X_t with kernel*

$$P_X^\pi(x' | x) = \sum_{a \in \mathcal{A}_1 \times \dots \times \mathcal{A}_n} \pi(a | x) P_X(x' | x, a).$$

3.2 Decentralized policies and partial observation

In a decentralized system, agent j need not observe the full augmented state. It may use

$$\pi_j(a_j | o_j(x), \zeta_j),$$

where $o_j(x)$ is its local observation and ζ_j is its controller memory. Once all controller memory is included in the augmented state, the induced joint action law is still a measurable function of the full certified state:

$$\pi(a | x) = \prod_{j=1}^n \pi_j(a_j | o_j(x), \zeta_j).$$

Thus partial observation does not invalidate the fixed-policy Markov-chain reduction. It only changes which policy class may be synthesized.

Proposition 1 (Fixed-policy reduction). *For any fixed decentralized joint policy profile π , the multi-agent Markov game over augmented state X induces a single Markov chain. Therefore, any Markov-chain certificate proof rule applicable to the induced kernel applies to the fixed policy profile.*

Proof. The augmented state contains all monitor memory and all policy memory. Given $X_t = x$ and a fixed policy profile, the distribution of local observations, joint actions, and the next augmented state is determined by $P_X^\pi(\cdot | x)$. The future is conditionally independent of the past given X_t , which is the Markov property. $\square$

3.3 Norm monitors and admission

A norm is admitted only if it can be compiled into a deterministic monitor and if its corresponding repair process has a provable bounded-response certificate. In the norm DSL, obligations have the form

$$\nu = \langle \text{Trigger}, \text{Post}, \tau \rangle,$$

meaning: if the trigger occurs, the postcondition must occur within τ steps. The deadline τ is a syntactic bound. The admission gate checks a semantic certificate: before the next discharge state, expected pending visits are bounded.

Assumption 1 (Admission gate). *For each admitted Streett pair (A_i, B_i) there is a deterministic monitor, a closed invariant region I , and a bounded nonnegative stopped-pair certificate satisfying the hypotheses of Theorem 1, uniformly over the supporting invariant and declared deviation class.*

Unbounded eventualities are not admitted into the active exported norm set unless they receive a separate recurrence proof. They may remain aspirational, diagnostic, or subject to future certification, but they do not receive exported certificate status.

4 Stopped-Pair Certificate Theory

The central formal object is not a global drift condition on all monitor states. It is a stopped-process statement for each Streett pair before the next discharge event.

Definition 2 (Stopped-pair progress certificate). *Let I be a closed invariant subset of the augmented state space. For a Streett pair (A, B) , let $\tau_B = \inf\{t \geq 0 : X_t \in B\}$. A function $W : I \rightarrow \mathbb{R}_{\geq 0}$ is a bounded stopped-pair progress certificate if there exist $\epsilon > 0$ and $W^{\max} < \infty$ such that $0 \leq W(x) \leq W^{\max}$ for all $x \in I$ and, for every $x \in I \setminus B$,*

$$\mathbb{E}[W(X_{t+1}) \mid X_t = x] \leq W(x) - \epsilon \mathbf{1}\{x \in A \setminus B\}. \quad (1)$$

No drift condition is imposed on states in B . The set B is the discharge/reset set for this pair.

Theorem 1 (Stopped-pair Streett certificate). *Suppose the fixed-policy certified process is a Markov chain on a closed invariant region I . If every Streett pair (A_i, B_i) has a bounded stopped-pair progress certificate W_i satisfying (1), then the induced process satisfies the Streett objective almost surely:*

$$\forall i, \quad X_t \in A_i \text{ i.o.} \implies X_t \in B_i \text{ i.o..}$$

Moreover, for each pair and each start state outside B_i ,

$$\mathbb{E}_x \left[\sum_{k < \tau_{B_i}} \mathbf{1}\{X_k \in A_i \setminus B_i\} \right] \leq \frac{W_i(x)}{\epsilon_i}. \quad (2)$$

Proof. Fix a pair (A, B) and write $\tau = \tau_B$. Define

$$M_t = W(X_{t \wedge \tau}) + \epsilon \sum_{k=0}^{t \wedge \tau - 1} \mathbf{1}\{X_k \in A \setminus B\}.$$

If $t \geq \tau$, the process is stopped. If $t < \tau$, then $X_t \notin B$ and (1) gives

$$\mathbb{E}[M_{t+1} \mid \mathcal{F}_t] \leq M_t.$$

Thus M_t is a nonnegative supermartingale. Taking expectations and using $W \geq 0$ gives

$$\epsilon \mathbb{E}_x \left[\sum_{k=0}^{t \wedge \tau - 1} \mathbf{1}\{X_k \in A \setminus B\} \right] \leq W(x).$$

Letting $t \rightarrow \infty$ and applying monotone convergence proves (2). Hence the number of pending $A \setminus B$ visits before the next B hit is finite almost surely.

Now consider the violation event that A occurs infinitely often but B only finitely often. This event is contained in the countable union over times m of the events

$$E_m = \{X_t \notin B \text{ for all } t \geq m\} \cap \{X_t \in A \setminus B \text{ infinitely often}\}.$$

(On the violation event there is a last B -visit at some time $m - 1$, possibly $m = 0$, and every A -visit after m is an $A \setminus B$ -visit.) Fix m . On $\{X_m \notin B\}$, apply (2) to the shifted process via the Markov property at time m and use $W \leq W^{\max}$:

$$\mathbb{E} \left[\sum_{k \geq m, k < \tau_B \circ \theta_m} \mathbf{1}\{X_k \in A \setminus B\} \mid \mathcal{F}_m \right] \leq \frac{W(X_m)}{\epsilon} \leq \frac{W^{\max}}{\epsilon} < \infty.$$

On E_m the shifted hitting time $\tau_B \circ \theta_m$ is infinite, so the left-hand sum counts all pending visits after m , which is infinite on E_m . A nonnegative random variable with finite conditional expectation is finite almost surely, so $\mathbb{P}(E_m) = 0$ for every m . Therefore the Streett violation event has probability zero. Repeating the argument for every pair gives the claim. $\square$

Remark 1 (Why no drift condition on B). *An infinite alternation between A and B satisfies the Streett condition. The certificate only needs to rule out infinite A -visits after B has stopped occurring. This is why the stopped proof is the appropriate proof rule and why B may reset or rebase the local certificate.*

Remark 2 (Sufficiency, not necessity). *The stopped-pair certificate is a sufficient proof rule. A Markov chain may satisfy the Streett condition without admitting this particular bounded drift certificate, especially when satisfaction follows from recurrence structure rather than monotone debt decrease. The exported guarantee of this paper is therefore certificate-relative: if the checker finds such certificates under the declared kernel and deviation class, the Streett objective follows. Where the simple stopped-pair rule fails on a genuinely satisfying chain, the complete certificate forms of Abate et al. [1] provide a strictly richer proof system at the cost of more checker machinery; substituting them changes only the local certificate object, not the glue, ledger, or composition layers built on top of it.*

Remark 3 (Repair debt is time-denominated). *Bound (2) reads $\mathbb{E}_x[\# \text{ pending visits before next } B_i] \leq W_i(x)/\epsilon_i$. The certified repair debt $d_i(x) = W_i(x)$ is therefore an upper bound, in units of ϵ_i , on the expected remaining pending duration of obligation i : the normalized quantity $W_i(x)/\epsilon_i$ has units of expected steps. This makes the price metaphor quantitative. It also suggests publishing the ϵ_i -normalized debt $\tilde{d}_i(x) = W_i(x)/\epsilon_i$ so that debts from norms with different strict-decrease margins are expressed in a common time currency and are directly comparable in the scarcity price q_e .*

The architecture uses the specification-local value-function construction of Abate et al. [2], rather than an absorbing-set construction, in abstraction lanes. Absorbing sets are fragile under abstraction; membership in A_i and B_i is monitor-local and survives monitor-preserving abstraction.

5 Certified Cognitive Glue

5.1 From certificate to price

For each module or interface e , let $\mathcal{P}(e)$ be the set of Streett pairs whose monitors depend on that interface. We publish a factored glue signal consisting of state-indexed prices and a history-indexed ledger,

$$g_t^e = \left((d_i(X_t), u_i(X_t))_{i \in \mathcal{P}(e)}, q_e(X_t), L_t^e \right),$$

where

$$d_i(x) = W_i(x), \quad \tilde{d}_i(x) = \frac{W_i(x)}{\epsilon_i}$$

are, respectively, raw certificate debt and normalized time-denominated repair debt. The normalized debt $\tilde{d}_i$ is the default quantity for comparing obligations with different drift margins; raw W_i may still be exported when a single certificate scale is being used. A minimal urgency signal is

$$\epsilon_i \mathbf{1}\{x \in A_i \setminus B_i\},$$

but implementations should publish a richer certified urgency price when the monitor provides deadline or slack data, for example

$$u_i(x) = \epsilon_i \mathbf{1}\{x \in A_i \setminus B_i\} + \kappa_i \frac{\mathbf{1}\{x \in A_i \setminus B_i\}}{1 + r_i(x)} + \eta_i [\epsilon_i - \widehat{\Delta W}_i(x)]_+.$$

Here $r_i(x)$ is remaining repair budget or deadline slack, and $\widehat{\Delta W}_i(x)$ is the checker-approved drift estimate. The default interface scarcity price is normalized by drift margin,

$$q_e(x) = \sum_{i \in \mathcal{P}(e)} \alpha_{e,i} \frac{W_i(x)}{\epsilon_i}, \quad \alpha_{e,i} \geq 0,$$

so that obligations are compared in a common expected-time currency. A raw-scale variant $\sum_i \alpha_{e,i} W_i(x)$ is allowed only when the exported glue contract declares why the participating debts are already commensurable. Finally, L_t^e is an append-only signed credit ledger of checker-verified changes $W_i(X_t) - W_i(X_{t+1})$ along declared monitor edges. Positive entries are credits and negative entries are debits. A positive-only cash ledger is not covered by the harvest lemmas below.

The point is not that agents understand the proof. The point is that the public coordination signal is reproducible from the proof object. Glue values are not self-reported by agents.

5.2 Glue locality

Dense global prices destroy compositionality. Therefore the glue is factored. A local module M_j may depend only on its local state and declared boundary glue.

Assumption 2 (Glue locality). *For each module M_j with internal state X^j and boundary state B^j , the local transition law satisfies*

$$P(X_{t+1}^j \mid X_t) = P(X_{t+1}^j \mid X_t^j, B_t^j).$$

Every glue component that affects M_j must be included in B^j .

In simulators, this assumption should be discharged by construction. In physical or mixed human-machine systems, it is a declared and audited factorization. If the assumption fails, local certificates do not compose. The system must enlarge the boundary, add an interface certificate, or verify the full product.

6 Potential-Based Shaping, Credit, and Attribution

The glue can be used as a training signal, but training is not verification. A policy trained with glue-derived rewards still has to pass the certificate checker. This section separates two mechanisms.

First, training shaping is a discounted reward transformation used inside reinforcement learning. Its purpose is to improve learning while preserving best-response incentives in the discounted game.

Second, transferable certificate credit is a public accounting object, such as a credit ledger, that may be used to compensate agents or explain which interface discharged an obligation. Transferable

credit must be signed, reset-safe, rebasing-safe, and attributed. Paying only for positive decreases of W is not safe.

Let Z_i denote the nonnegative certificate debt relevant to agent i ,

$$Z_i(x) = \sum_{k \in P_i} \lambda_{i,k} W_k(x), \quad \lambda_{i,k} \geq 0.$$

All statements below apply to a single W by taking $Z_i = W$, and to any nonnegative weighted aggregate by the displayed definition.

Definition 3 (Discounted shaping increment). *For discount factor $\gamma \in [0, 1)$ and potential $\Phi_i = -Z_i$, define*

$$F_i^\gamma(x, a, x') = \gamma \Phi_i(x') - \Phi_i(x) = Z_i(x) - \gamma Z_i(x').$$

This is the only shaping term used as a training reward.

Definition 4 (Signed certificate credit). *The transferable credit ledger records the signed, undiscounted certificate change*

$$C_i(x, x') = Z_i(x) - Z_i(x').$$

Positive entries are credits; negative entries are debits. A ledger that records only $\max\{C_i(x, x'), 0\}$ is a different mechanism and is not protected by the telescoping harvest bound.

Assumption 3 (Reset-safe accounting). *All monitor resets, norm discharges, episode counters, and credit balances that affect Z_i are included in the certified Markov state. A reset is therefore an ordinary transition $x \rightarrow x'$, and both $Z_i(x)$ and $Z_i(x')$ appear in the same shaping or credit increment. If an implementation terminates an episode and starts a fresh accounting episode, the new episode either begins exogenously or receives an explicit fresh-budget deposit recorded in the ledger. No agent-controllable reset may erase debits or allocate a new positive Z_i budget without a corresponding charge.*

Assumption 4 (Attribution-safe accounting). *Every signed ledger entry is booked to the agent, coalition, governance interface, or declared module edge responsible for the monitor transition. Positive credit and negative debit for the same accountable source must be recorded in the same accountable budget. If causation is ambiguous, the transition is assigned to a joint escrow account or treated as an unresolved externality, not paid as transferable credit. Side transfers among agents are outside the certificate unless represented in the certified state and ledger.*

Assumption 5 (Solvency and no-default accounting). *Transferable ledger credit is spendable only when backed: the corresponding debit is assigned to a budget that is collateralized, escrowed, or otherwise prevented from walking away from a negative balance. Any coalition whose members can make side transfers is represented as a single accountable budget for harvest analysis, or its internal transfers are included in the certified state. A debit assigned to an insolvent or disappearing account is treated as an unresolved externality rather than as an offset that licenses new spendable credit.*

Proposition 2 (Best-response invariance). *Fix the other agents' policies and an initial-state distribution. Suppose Z_i is bounded and the discounted objective uses the same γ as F_i^γ . Adding F_i^γ to agent i 's reward shifts every policy's infinite-horizon discounted value by the same constant. Hence it does not change the preference ordering over agent i 's policies and does not change its best-response set.*

Proof. For any finite horizon T ,

$$\sum_{t=0}^{T-1} \gamma^t F_i^\gamma(X_t, A_t, X_{t+1}) = \sum_{t=0}^{T-1} \gamma^t (\gamma \Phi_i(X_{t+1}) - \Phi_i(X_t)) = -\Phi_i(X_0) + \gamma^T \Phi_i(X_T).$$

Because $\Phi_i = -Z_i$ is bounded and $\gamma < 1$, the last term vanishes as $T \rightarrow \infty$. The infinite-horizon shaping contribution is $Z_i(X_0)$, which depends only on the initial state and not on the policy chosen after that state. $\square$

Remark 4 (Simultaneous shaping preserves equilibria). *The invariance holds for each agent separately at every fixed profile of the other agents' policies. Hence, if every agent is shaped simultaneously with its own potential $\Phi_i = -Z_i$, every agent's best-response correspondence is unchanged pointwise, and the set of Nash equilibria of the discounted shaped game equals that of the unshaped game. Shaping changes learning trajectories, not equilibrium structure.*

Lemma 1 (Discounted shaping harvest bound). *For any trajectory and any finite T ,*

$$\sum_{t=0}^{T-1} \gamma^t F_i^\gamma(X_t, A_t, X_{t+1}) = Z_i(X_0) - \gamma^T Z_i(X_T) \leq Z_i(X_0).$$

If Z_i is bounded, the infinite-horizon discounted shaping return equals $Z_i(X_0)$. Thus an agent cannot extract unbounded discounted shaping reward by repeatedly creating and resolving certificate debt; the entire discounted shaping account is capped by the initial debt budget.

Proof. Substitute $\Phi_i = -Z_i$ into the telescoping identity of the previous proof. Nonnegativity of Z_i gives the finite-horizon upper bound; boundedness gives $\gamma^T Z_i(X_T) \rightarrow 0$ pathwise, hence equality in the infinite-horizon limit. $\square$

Lemma 2 (Signed ledger harvest bound). *For any trajectory and finite T ,*

$$\sum_{t=0}^{T-1} C_i(X_t, X_{t+1}) = Z_i(X_0) - Z_i(X_T) \leq Z_i(X_0).$$

If a trajectory segment is a closed cycle in the certified state, $X_T = X_0$, then the signed certificate credit over the cycle is exactly zero. Under reset-safe, attribution-safe, and solvency/no-default accounting, repeated cycling, agent-triggered monitor resets, cross-agent debt creation and repair, ambiguous causation, and budget default cannot generate positive net extractable ledger surplus beyond the initial debt plus explicit exogenous deposits.

Proof. The sum telescopes directly:

$$\sum_{t=0}^{T-1} (Z_i(X_t) - Z_i(X_{t+1})) = Z_i(X_0) - Z_i(X_T).$$

The inequality follows from $Z_i \geq 0$. Reset-safe accounting prevents resets from silently replacing Z_i or erasing debits. Attribution-safe accounting prevents a positive repair entry from becoming transferable credit for one accountable budget while the corresponding debt-creation debit is hidden in another uncharged budget. Solvency/no-default accounting prevents the debit side from being assigned to an account that can disappear, default, or remain permanently negative while another account cashes out. Ambiguous entries remain in escrow and therefore are not paid as transferable credit. $\square$

The telescoping identity above controls the aggregate certificate change. It does not, by itself, control the economic quantity exposed to default: gross positive payouts can be legitimate compensation when matched by collectible debits or collateral. The object that must be bounded is therefore *net extractable surplus*, not the raw sum of positive payments. The next lemma states the budget-space conservation rule in that form.

Lemma 3 (Net ledger conservation across budgets). *Let $\mathcal{B}$ be the set of accountable budgets (agents, coalitions, declared module edges, escrow, governance). Suppose every checker-verified ledger entry from a transition $X_t \rightarrow X_{t+1}$ is booked to exactly one budget $b \in \mathcal{B}$, the signed entries booked to b sum to its balance bal_b , and Assumptions 3–5 hold. Let $D_T \geq 0$ denote cumulative explicit exogenous and governance deposits up to time T . For any accountable non-governance scope $S \subseteq \mathcal{B} \setminus \{\text{gov}\}$ that is closed under side transfers, define its settled surplus as*

$$\text{Surp}_S(T) = \text{Withdrawals}_S^+(T) - \text{CollectibleDebits}_S(T) - \text{CollateralLocks}_S(T) - \text{EscrowCharges}_S(T),$$

where withdrawals are gross positive spendable payouts and the subtracted terms are the collectible or locked resources that fund them. Then

$$\text{Surp}_S(T) \leq \sum_{b \in S} \text{bal}_b(T) + D_S(T),$$

where $D_S(T)$ is the portion of exogenous/governance deposits explicitly assigned to S . In particular, for the whole non-governance scope,

$$\text{Surp}_{\text{ng}}(T) \leq Z(X_0) - Z(X_T) + D_T \leq Z(X_0) + D_T.$$

On a closed certified cycle $X_T = X_0$ with no exogenous or governance deposits, the net extractable surplus of agents and coalitions is at most zero. Gross compensation may still occur inside the cycle, but it must be funded by a collectible debit, collateral lock, or escrowed resource; an uncollectible/defaulted debit cannot license spendable repair credit.

Proof. Partition the global telescoping sum of Lemma 2 by the budget to which each entry is booked:

$$Z(X_0) - Z(X_T) = \sum_{b \in \mathcal{B}} \left(\sum_{t: \text{entry booked to } b} (Z(X_t) - Z(X_{t+1})) \right) = \sum_{b \in \mathcal{B}} \text{bal}_b.$$

Explicit deposits add D_T to the available settlement mass. By attribution-safety, every spendable credit is assigned to a declared budget or scope. By solvency/no-default accounting, a gross positive payout is spendable only when its funding debit is collectible in the same accountable scope, collateralized, or locked in escrow. The funding debit/collateral/escrow term is therefore subtracted in $\text{Surp}_S(T)$. If the debit is assigned to an insolvent or disappearing account, the putative offset is reclassified as an unresolved externality and the corresponding repair credit is not spendable. Hence the settled net surplus of any closed non-governance scope is bounded by its signed settled balance plus deposits explicitly assigned to it. Taking the whole non-governance scope and using the global telescoping identity gives the displayed bound. The cycle case sets $Z(X_T) = Z(X_0)$ and $D_T = 0$. $\square$

Remark 5 (What conservation adds over telescoping). *Lemma 2 is a statement in Z -space; Lemma 3 is the corresponding statement in budget space, and it is the one that the default attack targets. The two together license the claim, recorded in Appendix obligation T3, that default cannot mint transferable surplus: telescoping bounds the total, attribution assigns it, and solvency prevents an uncollectible debit from being spent against. The lemma does not forbid ordinary gross compensation, such as paying a repairer from another agent's posted collateral; it forbids the coalition from extracting positive net surplus after the funding debit or collateral is counted.*

Remark 6 (Discounted cycles are not literally nonpositive). *A finite discounted traversal of a closed cycle can have positive discounted shaping return, because discounting front-loads earlier terms. For a cycle of length L starting and ending at the same state x , the discounted shaping sum for one traversal is $(1 - \gamma^L)Z_i(x) \geq 0$. Repeating the cycle does not create unbounded return; the total discounted shaping return over infinitely many repetitions is still $Z_i(x)$. The correct claim is bounded harvest, not nonpositive discounted credit on every finite cycle.*

Remark 7 (Positive-only credit is unsafe). *The signed-ledger condition is essential. If an implementation pays $\max\{Z_i(X_t) - Z_i(X_{t+1}), 0\}$ while ignoring negative entries, then a two-state loop with $Z_i = 1 \rightarrow 0 \rightarrow 1 \rightarrow 0 \rightarrow \dots$ can collect a positive payment on every downward step and pay nothing on every upward step. Such a mechanism reintroduces stress harvesting. It is outside the theorem.*

Remark 8 (Escrow is conservative, not neutral). *Routing ambiguous causation to escrow protects the harvest bound but is itself incentive-relevant: agents may be discouraged from genuinely helpful repairs whose causation the monitor cannot attribute. Tightening the attribution map, by enriching the declared monitor edges, recovers payable credit. The design trade is therefore explicit: attribution resolution is purchased with monitor state, and unattributable repair is unpaid rather than unsafe.*

Remark 9 (Escrow release rule). *Escrow must not grow without a release rule, or the conservatism above compounds over time. We adopt the following checkable discipline. An escrowed entry is released to an accountable budget if, within the same governance epoch, later certified evidence resolves its attribution to that budget; otherwise, at epoch close it expires to the governance scope. Both branches are ordinary certified transitions, so escrow balances remain inside the certified state and Lemma 3 continues to apply: release never creates positive mass, it only relabels existing mass from escrow to a backed budget.*

7 Minimal Worked Example

This section gives the smallest example that exercises the paper’s main accounting distinctions. It is not a realistic environment; its purpose is to make the stopped-pair rule, the signed ledger, attribution, solvency, and governance rebasing visible in one finite monitor.

One pending obligation. Let the certified monitor have two states, G (good/discharged) and P (pending repair), with the Streett pair $A = \{P\}$, $B = \{G\}$: if P is visited infinitely often, G must be too. Under the fixed certified policy, from P the repair transition reaches G with probability $q \in (0, 1]$ and otherwise stays in P ; from G an exogenous trigger may create a new obligation, $G \rightarrow P$ with some probability α , but no stopped-pair drift condition is imposed on $G \in B$. Set $W(G) = 0$, $W(P) = 1$. For the only state outside B ,

$$\mathbb{E}[W(X_{t+1}) \mid X_t = P] = (1 - q) \cdot 1 + q \cdot 0 = 1 - q = W(P) - q,$$

so W is a bounded stopped-pair certificate with $\epsilon = q$, $W^{\max} = 1$. Theorem 1 gives $\mathbb{E}_P[\sum_{k < \tau_G} \mathbf{1}\{X_k = P\}] \leq 1/q$, which is tight for this geometric repair. Cycles $G \rightarrow P \rightarrow G \rightarrow \dots$ are permitted; they satisfy the Streett condition because G recurs. By Remark 3, the debt $W(P) = 1$ corresponds to an expected $1/q$ pending steps before discharge.

Signed ledger versus positive-only credit. Take transferable debt $Z = W$, so the two monitor edges carry $C(P, G) = 1$ and $C(G, P) = -1$. Every closed cycle $G \rightarrow P \rightarrow G$ has net signed credit $-1 + 1 = 0$. A positive-only mechanism would instead pay $\max\{C(G, P), 0\} + \max\{C(P, G), 0\} = 0 + 1 = 1$ per cycle, even though the certified state returned to its start: the minimal stress-harvesting pump. The signed ledger eliminates it at the system level (Lemma 2).

Attribution and solvency. Suppose agent a controls $G \rightarrow P$ and agent b controls $P \rightarrow G$. The debit -1 must be charged to a 's budget, a joint coalition budget, or escrow; b 's $+1$ repair credit is spendable only if that debit is collectible (Assumption 5). If a has posted collateral, then b may legitimately receive a gross payment of $+1$ from that collateral, while the joint scope $\{a, b\}$ has net surplus 0. If a is insolvent or disappears, the debit becomes an unresolved externality and b 's credit is not spendable. This is exactly the case separated by Lemma 3: the Z -space cycle still sums to zero, and the budget-space bound forbids the coalition from cashing repair credit against a defaulted debit.

Governance rebasing. Suppose governance raises the certified debt while pending, $Z_{\text{old}}(P) = 1 \rightarrow Z_{\text{new}}(P) = 2$. The rebasing amount $\Delta^{\text{gov}} = Z_{\text{new}}(P) - Z_{\text{old}}(P) = 1$ is booked as a governance deposit, not agent credit. If the next transition is $P \rightarrow G$ under the new certificate, the ledger entry is $+2$ on the new scale, but one unit came from the governance deposit; the discharging agent collects at most the originally collateralized unit, and the extra unit is a permissioned, governance-funded repair subsidy (see Remark 10). Closing the old ledger and opening the new one prevents repeated recertification from minting hidden credit.

8 Norm Learning as Certificate-Prior Induction

The architecture uses norm learning as a proposal mechanism. Each agent maintains a finite norm-state transducer, for example a binned log-odds vector over candidate norms. If $b_j(\nu)$ is agent j 's log-odds bin for norm ν , public evidence events e_t update it by a finite transition rule

$$b_{j,t+1}(\nu) = \text{clip}_{\mathcal{B}}(b_{j,t}(\nu) + \Delta_{\nu}(e_t, o_{j,t})),$$

where $\mathcal{B}$ is a finite bin set and Δ_{ν} is supplied by the norm-learning model. Other finite transducer designs can be substituted.

A candidate norm follows the lifecycle

$$\text{proposed} \rightarrow \text{estimated} \rightarrow \text{compiled} \rightarrow \text{admitted} \rightarrow \text{certified} \rightarrow \text{active}.$$

Estimated norms may guide exploration and planning. Only certified norms produce exported guarantees and public credit postings.

This distinction addresses a known problem in norm learning: obligations are harder to transmit than prohibitions because the discharge event may be rarely observed. In our architecture, verified obligation discharge is echoed into the public glue ledger. A novice need not witness the cleaner clean the river; it can observe that the certified cleaning obligation was discharged.

9 Humans, Governance, and Permissioned Hijacking

Humans are not modeled as certified norm-compliant planners. Instead, they enter in two ways.

First, humans can be placed inside a bounded deviation class D_H , such as an action mask, a logit-bounded perturbation around an interpretable nominal model, or a finite transducer. Robust claims are only as broad as this class.

Second, humans or human-authorized governance processes hold permissioned hijack rights over glue parameters. They may request tax-like or subsidy-like offsets, norm vetoes, or changes to admission thresholds. Such operations are themselves transitions in the certified model. A requested operation must pass recertification or be rejected with a counterexample.

9.1 Ledger rebasing under governance changes

Governance changes can alter the certificate, the glue weights, the monitor, or the admitted norm set. Such changes break a naive telescoping identity unless the old ledger is closed and a new ledger is opened.

At a governance intervention time t , let Z_{old} be the debt used by the ledger before the intervention and Z_{new} the debt after recertification. The old ledger is closed at $Z_{\text{old}}(X_t)$, the new ledger opens at $Z_{\text{new}}(X_t)$, and the rebasing amount

$$\Delta_t^{\text{gov}} = Z_{\text{new}}(X_t) - Z_{\text{old}}(X_t)$$

is booked as a governance deposit or withdrawal. It is not credited to agents. Repeated recertification therefore cannot become an agent harvest channel unless the governance process itself injects explicit budget.

Remark 10 (Upward rebasing is a permissioned subsidy). *Raising Z while an obligation is pending, as in the worked example, deposits governance budget that the discharging agent will collect on top of the originally collateralized debt. Operationally this is a certified repair subsidy paid from the governance scope to whoever discharges, and it is a deliberate lever, not an accounting leak: it is bounded by Δ_t^{gov} , recorded in the rebasing log, and never sourced from another agent’s budget. Symmetrically, downward rebasing is a permissioned tax. The architecture certifies that these flows are governance-funded and logged; it does not certify that the governance choice to subsidize or tax is wise.*

This mirrors an economic distinction. Machines coordinate through prices. Humans choose which institutional objectives the price system should serve. The technical system can check whether a proposed intervention preserves a formal specification; it cannot certify the legitimacy of the human governance process itself.

10 Compositional Certification

The previous sections describe how to certify one augmented Markov chain. A realistic collective cannot usually be certified as one monolithic product state. We therefore need conditions under which local certificates can be combined. This section gives a deliberately strong but checkable composition theorem. The theorem does not discover the correct decomposition. It says that if all cross-module effects have been declared and if their gains are small enough, then a weighted sum of local certificate debts is itself a global certificate debt.

The key distinction is between real coupling and declared interface coupling. Local verification is sound only when every way in which one module can worsen another module’s certificate is represented in the interface budget. Undeclared couplings void the theorem.

Let $r = 1, \dots, R$ index certified local obligations or local Streett components. These indices may correspond to different modules, or to several obligations inside the same module. Let $Z_r \geq 0$ be the local certificate debt associated with component r , and let

$$H_r(x) \in \{0, 1\}$$

indicate that component r is currently in its pending, obligation-active region. Let $\epsilon_r > 0$ be the local strict-decrease margin available without interface disturbance. Define the dissipation vector

$$d_r(x) = \epsilon_r H_r(x), \quad d(x) = (d_1(x), \dots, d_R(x))^\top.$$

Assumption 6 (Interface-complete drift budget). *There exists a nonnegative matrix $\Gamma \in \mathbb{R}_{\geq 0}^{R \times R}$ with zero diagonal such that, for every certified global state x and every component r ,*

$$\mathbb{E}[Z_r(X_{t+1}) - Z_r(X_t) \mid X_t = x] \leq -d_r(x) + \sum_{q=1}^R \Gamma_{rq} d_q(x).$$

The term $\Gamma_{rq} d_q$ is the maximum amount of component q 's active obligation budget that may be spent worsening component r through declared interfaces. All such effects must be included in the declared interface graph and in the certified state.

Assumption 7 (Strict small gain). *The gain matrix satisfies $\rho(\Gamma) < 1$, where ρ denotes spectral radius.*

Theorem 2 (Linear small-gain composition). *Under the interface-complete drift budget and strict small-gain assumptions, define*

$$\lambda = (I - \Gamma^\top)^{-1} \mathbf{1}.$$

Then $\lambda > 0$ and

$$Z_{\text{glob}}(x) = \sum_{r=1}^R \lambda_r Z_r(x)$$

satisfies

$$\mathbb{E}[Z_{\text{glob}}(X_{t+1}) - Z_{\text{glob}}(X_t) \mid X_t = x] \leq -\mathbf{1}^\top d(x).$$

Consequently the global debt has strict negative drift whenever at least one local obligation is active, and nonpositive drift whenever none is active.

Proof. Because $\Gamma \geq 0$ and $\rho(\Gamma) < 1$, the inverse $(I - \Gamma^\top)^{-1}$ exists and is nonnegative. Thus $\lambda > 0$ and

$$\lambda^\top (I - \Gamma) = \mathbf{1}^\top.$$

Multiplying the componentwise drift inequalities by λ_r and summing over r gives

$$\mathbb{E}[\Delta Z_{\text{glob}} \mid x] \leq \lambda^\top (-d(x) + \Gamma d(x)) = -\lambda^\top (I - \Gamma) d(x) = -\mathbf{1}^\top d(x).$$

□

Remark 11 (Idle-module caveat). *The theorem applies only when all cross-module harm is charged to declared active interface budgets. If an idle module can perturb a neighbor while having no active dissipation budget of its own, that perturbation must be represented as standing debt, affine disturbance, or undeclared coupling. Otherwise Theorem 2 does not apply.*

Corollary 1 (Global liveness). *In the setting of Theorem 2, suppose additionally that each Z_r is bounded on the certified invariant region, so that $Z_{\text{glob}} \leq \sum_r \lambda_r Z_r^{\max} =: Z_{\text{glob}}^{\max}$, and that $H_r = 0$ means component r is either discharged/inactive according to its certified monitor rather than silently pending outside the indicator. Let*

$$A_{\text{glob}} = \{x : \exists r, H_r(x) = 1\}, \quad B_{\text{glob}} = \{x : \forall r, H_r(x) = 0\}, \quad \epsilon_{\text{glob}} = \min_r \epsilon_r.$$

Then Theorem 2 gives, for every $x \in I \setminus B_{\text{glob}} = A_{\text{glob}}$,

$$\mathbb{E}[Z_{\text{glob}}(X_{t+1}) \mid X_t = x] \leq Z_{\text{glob}}(x) - \epsilon_{\text{glob}},$$

so Z_{glob} is a bounded stopped-pair progress certificate for the aggregate global pair $(A_{\text{glob}}, B_{\text{glob}})$. By Theorem 1, the composed system almost surely cannot keep some declared pending component active infinitely often without infinitely often returning to the all-discharged/inactive region, and from any state the expected aggregate pending occupancy before the next all-discharged/inactive hit is at most $Z_{\text{glob}}(x)/\epsilon_{\text{glob}}$. This corollary exports the aggregate bounded-response monitor; local Streett guarantees still require the local monitors' own stopped-pair certificates or an explicit proof that exiting $H_r = 1$ coincides with discharge for component r .

Proof. On A_{glob} at least one $d_r(x) = \epsilon_r H_r(x)$ is positive, so $\mathbf{1}^\top d(x) \geq \min_r \epsilon_r = \epsilon_{\text{glob}}$, and the drift bound of Theorem 2 gives the displayed inequality. On B_{glob} , $d(x) = 0$ and the drift is nonpositive, though the stopped-pair rule does not require this. Boundedness and nonnegativity of Z_{glob} follow from those of the Z_r and $\lambda > 0$. The additional monitor-semantics assumption ensures that B_{glob} is really the all-discharged/inactive region, not an artifact of a missing pending indicator. Theorem 1 then applies verbatim with $W = Z_{\text{glob}}$, $\epsilon = \epsilon_{\text{glob}}$. $\square$

Corollary 2 (Debt-contraction composition). *Suppose there exist $a_r > 0$ and a nonnegative off-diagonal matrix G such that*

$$\mathbb{E}[Z_r(X_{t+1}) - Z_r(X_t) \mid X_t = x] \leq -a_r Z_r(x) + \sum_{q=1}^R G_{rq} Z_q(x)$$

for every r and every certified state x . If $\rho(\text{diag}(a)^{-1}G) < 1$, then with

$$\lambda = (\text{diag}(a) - G^\top)^{-1} \mathbf{1}$$

the weighted aggregate $Z_{\text{glob}} = \sum_r \lambda_r Z_r$ satisfies

$$\mathbb{E}[Z_{\text{glob}}(X_{t+1}) - Z_{\text{glob}}(X_t) \mid X_t = x] \leq -\mathbf{1}^\top Z(x),$$

where $Z(x) = (Z_1(x), \dots, Z_R(x))^\top$. More generally, choosing a different positive right-hand vector in the M -matrix solve gives positive margins $\eta_r > 0$ with right-hand side $-\sum_r \eta_r Z_r(x)$.

Proof. Write $A = \text{diag}(a)$. Since $\rho(A^{-1}G) < 1$ with $A^{-1}G \geq 0$, the matrix $A - G$ is a nonsingular M -matrix, and so is its transpose $A - G^\top$; both have nonnegative inverses. Then

$$(A - G^\top)^{-1} = A^{-1}(I - G^\top A^{-1})^{-1} = A^{-1} \sum_{k \geq 0} (G^\top A^{-1})^k \geq A^{-1},$$

entrywise, so $\lambda = (A - G^\top)^{-1} \mathbf{1} \geq A^{-1} \mathbf{1} > 0$. Moreover $\lambda^\top (A - G) = \mathbf{1}^\top ((A - G^\top)^{-1})^\top (A - G) = \mathbf{1}^\top (A - G)^{-1} (A - G) = \mathbf{1}^\top$, using $(A - G^\top)^\top = A - G$. Multiplying the r -th drift inequality by λ_r and summing gives

$$\mathbb{E}[\Delta Z_{\text{glob}} \mid x] \leq -\lambda^\top (A - G) Z(x) = -\mathbf{1}^\top Z(x).$$

$\square$

Proposition 3 (Affine-disturbance small gain). *Suppose the local drift inequalities take the form*

$$\mathbb{E}[Z_r(X_{t+1}) - Z_r(X_t) \mid X_t = x] \leq -d_r(x) + \sum_{q=1}^R \Gamma_{rq} d_q(x) + b_r,$$

where $b_r \geq 0$ is a declared disturbance budget and $\rho(\Gamma) < 1$. With $\lambda = (I - \Gamma^\top)^{-1} \mathbf{1}$,

$$\mathbb{E}[\Delta Z_{\text{glob}} \mid x] \leq -\mathbf{1}^\top d(x) + \lambda^\top b.$$

Thus the aggregate debt strictly improves whenever total active repair pressure exceeds the weighted disturbance budget: $\mathbf{1}^\top d(x) > \lambda^\top b$.

Proof. Multiply the r -th inequality by λ_r and sum. The drift terms contribute $-\lambda^\top (I - \Gamma) d(x) = -\mathbf{1}^\top d(x)$ exactly as in Theorem 2; the disturbances contribute $\lambda^\top b$. $\square$

Remark 12 (What the affine variant certifies). *The affine bound is an input-to-state style state-ment, not a strict certificate. It implies that Z_{glob} drifts downward only outside a sublevel neighborhood determined by the disturbance budget; pending obligations may persist within that neighborhood. To export a stopped-pair guarantee under disturbance, either shrink b so that $\epsilon_{\text{glob}} > \lambda^\top b$ holds with margin (then Corollary 1 applies with the reduced margin $\epsilon_{\text{glob}} - \lambda^\top b$), or treat the disturbance as a declared deviation class and recertify.*

Remark 13 (How to read the gain matrix). *The entry Γ_{rq} is not a correlation coefficient and not a learned attention weight. It is a certified worst-case interface budget: when obligation q is active, how much can its repair dynamics worsen component r 's debt through the declared glue interface? If the checker later finds a transition that worsens Z_r but is not covered by any declared $\Gamma_{rq} d_q$, the result is not a theorem failure; it is a decomposition failure. The missing coupling must be added to the interface graph, the abstraction must be split, or the full product must be verified.*

Remark 14 (Interface price maps). *Factored glue price maps may be used inside the interface budgets only through certified Lipschitz constants. If a price map from component q to component r has Lipschitz constant L_{rq} and the local transition sensitivity of Z_r to that price is S_{rq} , then one may set $\Gamma_{rq} \geq S_{rq} L_{rq}$ after accounting for the units of the dissipation budget. The small-gain test is then a design rule: choose price-map gains so that $\rho(\Gamma) < 1$.*

11 Least-Forgetting Abstraction

Scalability is the main obstacle. The exact finite lane is only a smoke test because the product state grows too quickly. The primary scalable regime uses abstraction.

An abstraction can fail by merging concrete states that require different repair actions. We use a certificate-aware forgetting cost to prioritize refinement. The score is always defined relative to a declared sampling measure. Let ρ be either the certified nominal rollout distribution ρ_{nom} or a counterexample-guided stress distribution ρ_{stress} . For an abstraction α with cells C , define

$$K_\rho(\alpha) = \sum_{C \in \alpha} \left[\min_d \sum_{x \in C} \mu_\rho(x) \ell(d, x) - \sum_{x \in C} \mu_\rho(x) \min_d \ell(d, x) \right],$$

where d ranges over abstract repair modes, $\ell(d, x)$ is local certificate-relevant loss, and $\mu_\rho(x)$ is the empirical or certified state weight induced by ρ . We report both

$$K_{\rho_{\text{nom}}}(\alpha) \quad \text{and} \quad K_{\rho_{\text{stress}}}(\alpha)$$

when stress trajectories are available.

A risk-sensitive version treats the cell-aliasing loss L_α as a random variable under the same trajectory generator and sets

$$K_\rho^{\text{risk}}(\alpha) = \mathbb{E}_\rho[L_\alpha] + \beta \text{CVaR}_{0.95, \rho}(L_\alpha).$$

This score is not a proof. Estimated K_ρ from sampled trajectories can miss rare structured failures. It is used only to prioritize counterexample-guided abstraction refinement. A formal abstraction claim requires an independent robust check over the abstract cell, using interval, symbolic, exhaustive, or parameter-coupled bounds.

12 Missing Markets and Counterexamples

In economics, an externality can be understood as a relevant effect that is not priced. In this architecture, a missing market is a transition that increases certificate risk without moving any declared glue component, monitor variable, or accountable ledger edge.

The checker therefore supplies counterexamples that become design obligations. If a state-action transition violates the stopped-pair inequality, small-gain budget, attribution rule, or rebasing rule, the system must either add a monitor variable, add a glue component, refine the abstraction, route the transition to escrow, or reject the policy. This is a counterexample-guided missing-markets ledger.

13 Experimental Program

The first empirical test should be deliberately small. A suitable environment is a two-agent or three-agent commons game with resource harvesting, pollution, compensation, and limited sanctions. Candidate norms include: do not steal; clean pollution within a deadline; compensate cleaners within a deadline; repair after violation; do not sanction after repair.

Baselines. Compare: (i) no norms, (ii) norm learning without certification, (iii) reward shaping without certificate-derived prices, (iv) certificate checking without public glue, and (v) certified cognitive glue.

Primary tests.

1. **Certificate pass rate:** how often does the trained joint policy admit valid bounded stopped-pair certificates?
2. **Adversarial attribution and default test:** one agent creates certificate debt, another resolves it, and a coalition attempts to extract positive net surplus through side transfers, budget default, or ambiguous responsibility. Ambiguous causation must route to escrow rather than payout; a defaulting debited budget must be treated as an unresolved externality so that the matching repair credit is not spendable (Lemma 3).
3. **Governance rebasing attack:** repeated parameter changes and recertifications are attempted as a credit-creation channel. The rebasing ledger should charge all certificate discontinuities to governance, not agents.

4. **Obligation transmission:** do novices learn obligations faster when discharge events are posted to the public ledger?
5. **Missing-market closure:** can seeded unpriced harms be found by counterexamples and closed by adding monitor or glue components?
6. **Human hijack demo:** can a subsidy-like intervention pass recertification, while an unsafe intervention is rejected with an interpretable counterexample?

Artifacts. The output should include a JSON certificate containing the environment hash, policy hash, monitor, admitted norms, value tables or bounds, glue map, Lipschitz budgets, deviation class, attribution ledger, solvency/collateral records, governance rebasing log, data provenance hash, and checker status.

14 Limitations

The proposal does not solve open-ended alignment. It certifies satisfaction of formal specifications under modeled dynamics and declared deviation classes. If the specification is wrong, the certificate proves the wrong thing. If humans behave outside D_H , the human robustness claim does not apply. If sensors poison the model used to compute W , the certificate is only as trustworthy as the provenance and robust set allow.

The exact finite lane will not scale to realistic collectives. Abstraction is necessary, and composition is available only under the declared-interface small-gain theorem of Section 10. Least-forgetting scores are useful diagnostics but not proof substitutes. Potential-based shaping preserves discounted policy preferences but does not make an unsafe learned policy safe; the checker remains authoritative.

Finally, permissioned hijack rights introduce governance risk. The architecture can reject interventions that break formal certificates, and it can prevent governance rebasing from becoming agent credit, but it cannot certify that the humans choosing objectives are legitimate or wise.

15 Conclusion

Certified cognitive glue turns a cooperation certificate into the public coordination medium of a multi-agent system. The same function W that proves progress toward obligation discharge also defines repair debt, urgency, scarcity, and signed attributed credit. This gives a precise analogue of prices for artificial collectives: not a hand-designed signal that the verifier later audits, but a sparse public signal derived from the proof object itself.

The design brings together three traditions: supermartingale verification, norm learning in Markov games, and economic accounts of coordination through shared scarcity signals. The stopped-pair theorem states the exact Streett obligation that is certified; the signed attribution and rebasing rules state the exact accounting condition needed for transferable credit; and the composition result shows one way this can scale when cross-module disturbances are declared and contractive. Its strongest immediate use is modest: small, auditable cooperative systems with bounded obligations and explicit human governance hooks. Its longer-term promise is a path from empirical cooperation to public, machine-checkable institutional certificates.

A Open Theorem Obligations

T1: Value-function certificate.

Imported from the value-function supermartingale result under the theorem hypotheses. Local obligation: discharge admission-gate hypotheses and boundedness/integrability checks.

T2: Stopped-pair Streett theorem.

Closed by Theorem 1 for bounded certificates on the certified invariant region.

T3: Harvest bounds.

Proved in Section 6. Discounted potential shaping is best-response invariant and bounded by initial certificate debt. Transferable credit is harvest-safe only under signed, reset-safe, attribution-safe, solvency/no-default, and rebasing-safe accounting; budget-space conservation is closed by Lemma 3.

T4: Abstraction transfer.

If a monitor-preserving abstraction passes the robust stopped-pair and small-gain checks under parameter-coupled uncertainty, prove the concrete process satisfies the Streett objective.

T5: Composition.

Closed for the linear declared-interface case by Theorem 2, the contraction corollary, and the global-liveness export of Corollary 1. The affine-disturbance variant gives a bounded-risk statement, not a full strict certificate. Automatic decomposition, hidden couplings, nonlinear gain discovery, idle-module disturbances, and arbitrary factored POMDP-style composition remain open.

B Related Work

This proposal is closest to four lines of work. First, it uses supermartingale certificates for stochastic omega-regular verification, including recent complete certificate frameworks for ω -regular properties [1], and the recent connection between value functions and certificates [2]. Second, it uses norm-augmented Markov games and Bayesian rule induction as a source of candidate normative systems [3]. Third, it borrows from economic and biological accounts of cognitive glue, especially the view that price systems coordinate plans by publishing scarcity information [4]. Fourth, it adopts potential-based shaping invariance as a way to use certificate values during training without changing best-response rankings [5].

The least-forgetting abstraction discipline is motivated by work on weighted forgetting cost and risk-sensitive aliasing in recurrent systems [6, 7]. In this paper, these scores are not used as formal proof conditions; they are used to guide refinement before a hard checker supplies the guarantee.

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